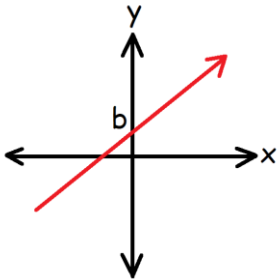
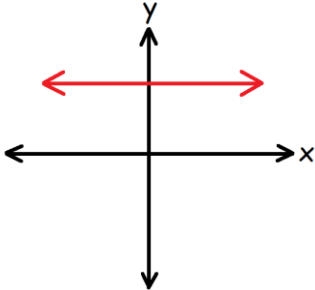
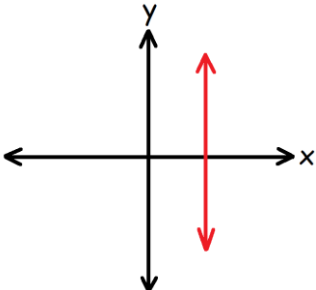
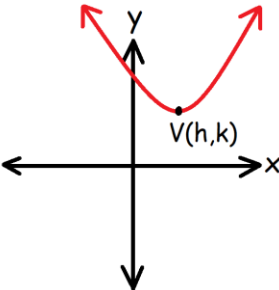
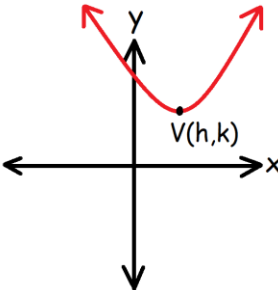
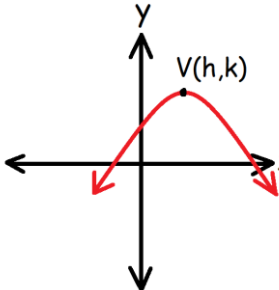
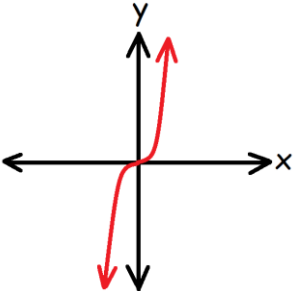
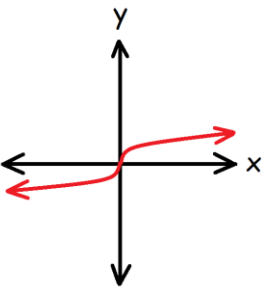
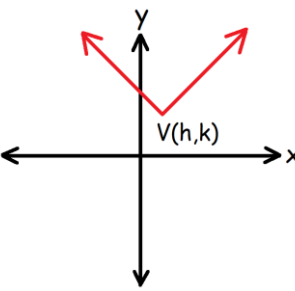
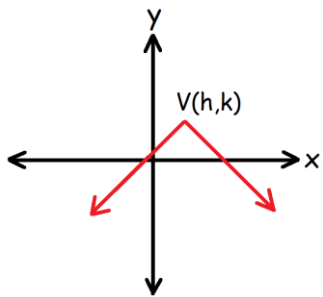
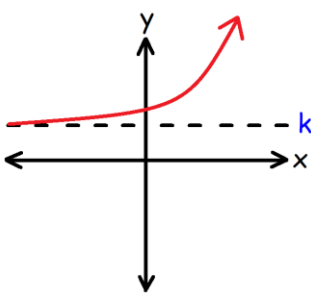
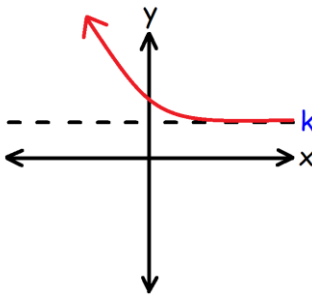
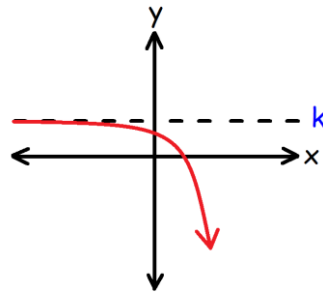
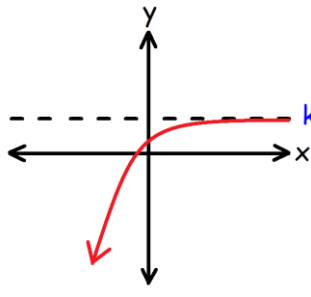
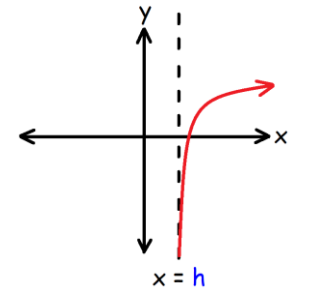
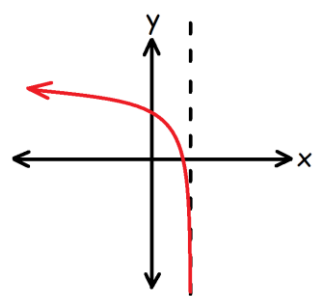
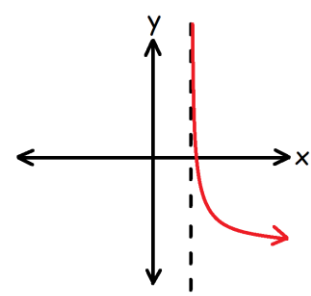
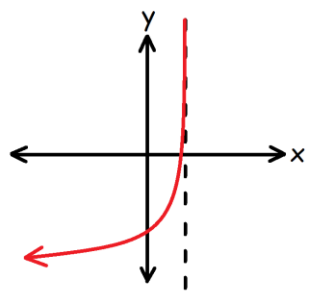
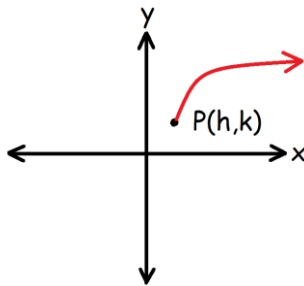
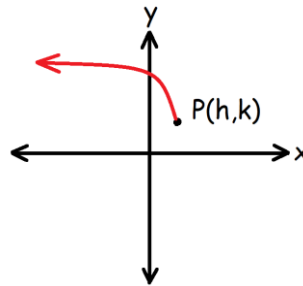
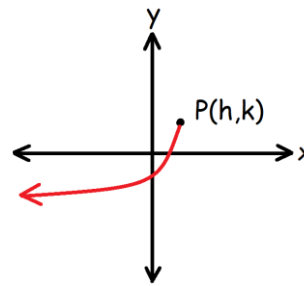
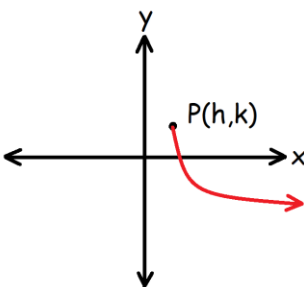
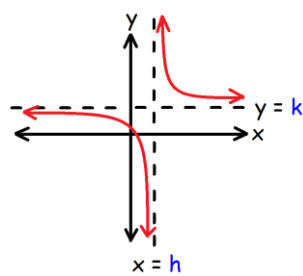
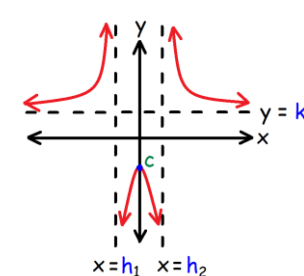
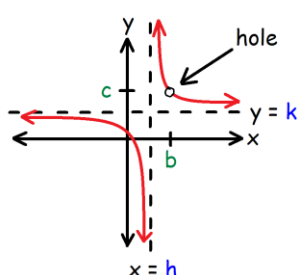
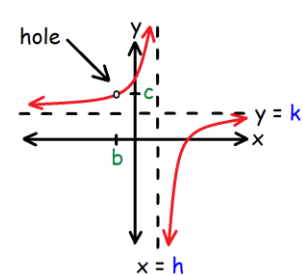
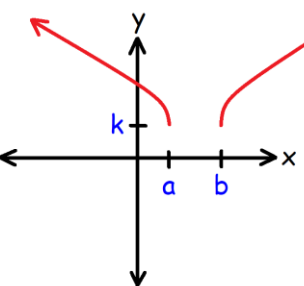


Domain and Range Formula Sheet:

Linear Functions: $y = mx + b$  Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$	Horizontal Line: (m = 0) $y = k$  Domain: $(-\infty, \infty)$ Range: $\{k\}$	Vertical Line: (m = undefined) $x = h$  Domain: $\{h\}$ Range: $(-\infty, \infty)$
Quadratic Functions: (a = +) $y = ax^2 + bx + c$  Domain: $(-\infty, \infty)$ Range: $[k, \infty)$ $h = -b/2a$ $k = f(-b/2a)$	Quadratic Functions: (a = +) $y = a(x - h)^2 + k$  Domain: $(-\infty, \infty)$ Range: $[k, \infty)$ Vertex Form	Quadratic Functions: (a = -) $y = ax^2 + bx + c$  Domain: $(-\infty, \infty)$ Range: $(-\infty, k]$ Standard Form
Cubic Functions: $y = ax^3 + bx^2 + cx + d$  Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$	Cube Root Function: $y = \sqrt[3]{x}$  Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$	Absolute Value Functions: (a = +) $y = a x - h + k$  Domain: $(-\infty, \infty)$ Range: $[k, \infty)$

Absolute Value Functions: ($a = -$) $y = a x - h + k$  Domain: $(-\infty, \infty)$ Range: $(-\infty, k]$	Exponential Functions: $y = a^x + k$  Domain: $(-\infty, \infty)$ Range: (k, ∞)	Exponential Functions: $y = a^{-x} + k$  Domain: $(-\infty, \infty)$ Range: (k, ∞)
Exponential Functions: $y = -a^x + k$  Domain: $(-\infty, \infty)$ Range: $(-\infty, k)$	Exponential Functions: $y = -a^{-x} + k$  Domain: $(-\infty, \infty)$ Range: $(-\infty, k)$	Logarithmic Functions: $y = \log_a(x - h)$  Domain: (h, ∞) Range: $(-\infty, \infty)$
Logarithmic Functions: $y = \log_a(h - x)$  Domain: $(-\infty, h)$ Range: $(-\infty, \infty)$	Logarithmic Functions: $y = -\log_a(x - h)$  Domain: (h, ∞) Range: $(-\infty, \infty)$	Logarithmic Functions: $y = -\log_a(h - x)$  Domain: $(-\infty, h)$ Range: $(-\infty, \infty)$

Radical Functions: $y = \sqrt{x - h} + k$  Domain: $[h, \infty)$ Range: $[k, \infty)$	Radical Functions: $y = \sqrt{h - x} + k$  Domain: $(-\infty, h]$ Range: $[k, \infty)$	Radical Functions: $y = -\sqrt{h - x} + k$  Domain: $(-\infty, h]$ Range: $(-\infty, k]$
Radical Functions: $y = -\sqrt{x - h} + k$  Domain: $[h, \infty)$ Range: $(-\infty, k]$	Rational Functions: ($a > 0$) $y = \frac{a}{x - h} + k$  Domain: $(-\infty, h) \cup (h, \infty)$ Range: $(-\infty, k) \cup (k, \infty)$	Rational Functions: ($a > 0, k > c$) $y = \frac{a}{(x - h_1)(x - h_2)} + k$  D: $(-\infty, h_1) \cup (h_1, h_2) \cup (h_2, \infty)$ R: $(-\infty, c) \cup (k, \infty)$
Rational Functions: $y = \frac{a(x - b)}{(x - h)(x - b)} + k$  D: $(-\infty, h) \cup (h, b) \cup (b, \infty)$ R: $(-\infty, k) \cup (k, c) \cup (c, \infty)$ Note: $b > h$ and $c > k$.	Rational Functions: $y = \frac{a(x - b)}{(x - h)(x - b)} + k$  D: $(-\infty, b) \cup (b, h) \cup (h, \infty)$ R: $(-\infty, k) \cup (k, c) \cup (c, \infty)$ Note: $h > b$ and $c > k$.	Complex Radical Functions: $y = \sqrt{(x - a)(x - b)} + k$  Domain: $(-\infty, a] \cup [b, \infty)$ Range: $[k, \infty)$

Steps for Finding the Domain of Certain Functions:

1. For $y = \frac{1}{ax + b} \rightarrow$ Set $ax + b \neq 0$ and solve for x .

2. For $y = \sqrt{ax + b} \rightarrow$ Set $ax + b \geq 0$ and solve for x .

3. For $y = \sqrt{ax^2 + bx + c} \rightarrow$ Set $ax^2 + bx + c \geq 0$, factor and solve for x .

4. For $y = \log_a(bx + c) \rightarrow$ Set $bx + c > 0$ and solve for x .

5. For $y = \frac{1}{\sqrt{ax + b}} \rightarrow$ Set $ax + b > 0$ and solve for x .

6. For $y = \frac{1}{\sqrt{ax^2 + bx + c}} \rightarrow$ Set $ax^2 + bx + c > 0$, factor and solve for x .

Note: A number line may be helpful for #3 and #6.